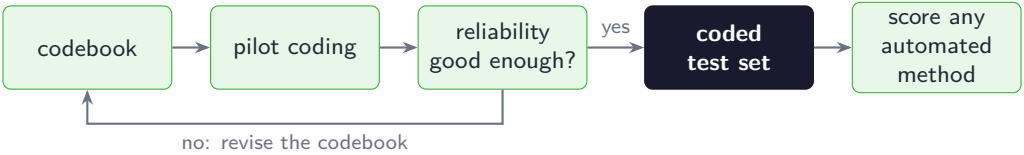
Pixels, Frames & Transformers

Coding Images and Classic Classifiers · Module 04

Denise J. Roth, PhD Candidate
Strategic Communication Group · Wageningen University & Research

Human Coding Is the Benchmark

Every automated measure in this course, from a brightness score to a vision-language model, is only as good as its agreement with careful human judgment.



In this module you code 40 images. Those codes become the **test set** for every method in Parts 1 to 3, so all of them are scored on the same images.

Choosing the Unit of Analysis

The same picture can be coded at several levels. Each level answers different questions.

The whole image
Does this photo show a protest?

Each person
How are women shown in news photos?

A region
What is in the foreground? How much of the package is the mascot?

The post
Does the caption change what the image means?

This module codes whole images. Regions and people return in modules 12 and 14, posts in module 11.

A Short Visual Codebook

	"yes" when ...	"no" when ...
protest	people engaged in protest are visible: marching, gathering with signs, a blockade, a sit-in	only objects (a poster, an empty square), police alone, or a crowd with no sign of protest
crowd	roughly ten or more people are visible, including the background	fewer than about ten people
police	officers, police vehicles or police barriers are visible	no police (stewards don't count)
signs	a sign, banner or flag with a message is carried or displayed by people	no signs, or only street, shop and traffic signs

Code what you can see, not what you know about the event. If you can't tell, choose the answer that needs less inference, and write down the image: it probably calls for a new rule.

Coding in Practice

Tools

In the notebook: buttons under each image, codes saved after every click. Quick, for whole-image codes and one coder.

Label Studio (open source): boxes and regions, several coders, a web interface. Worth setting up for a real project.

A spreadsheet with file names: the fall-back that always works.

Habits

- **Code blind**: don't let coders see where an image came from
- **Shuffle** the order, and code in short sessions
- **Write down new rules** as hard cases come up
- **Recode** a sample later: is your coding *stable*?
- Add a **second coder**: is it *reproducible*? This is the standard for publication.

Measuring Agreement

- **Percent agreement**: share of images coded the same way; rewards luck
- **Cohen's kappa**: corrected for chance, $\kappa = (p_o - p_e) \,/\, (1 - p_e)$
- **Krippendorff's alpha**: any number of coders, missing codes; the standard in content analysis

$\alpha \geq .800$: reliable .667–.800: tentative conclusions only (Krippendorff, 2018)

Why percent agreement misleads

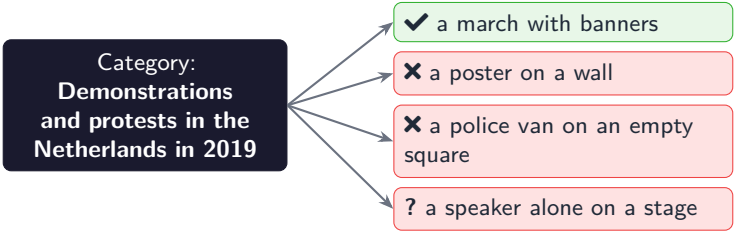
Police appears in 5% of images. One coder looks carefully; the other always says "no" without looking.

percent agreement	**0.95**
Cohen's kappa	**0.00**

Report a chance-corrected measure. Rare categories need more images before you can judge their reliability at all.

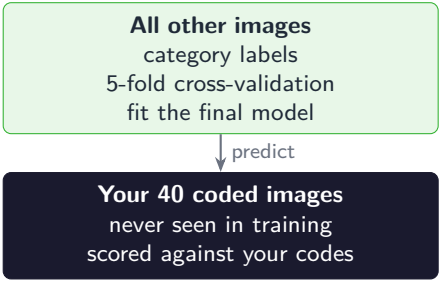
Free Labels Are Noisy Labels

Every image comes with a label you didn't code: the **category** it was filed under. Metadata, hashtags, search terms and accounts work the same way.



A category says **where an image was filed**, not **what it shows**. Train on free labels if nothing better exists, but always evaluate against human codes.

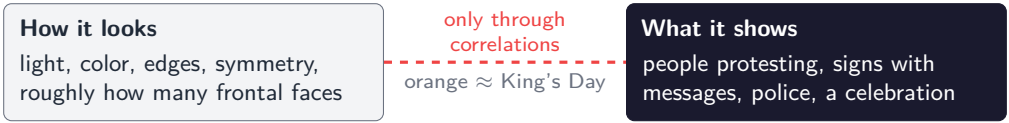
A Classic Classifier, Honestly Evaluated



- Input: the 11 features from module 03
- **Logistic regression**: one weight per feature
- **Random forest**: many trees; thresholds and interactions
- Unequal classes: balanced class weights; report **balanced accuracy** (0.5 = chance) and **F1** per class

A model scored on images it was trained on looks far better than it is. Keep the test set out of training and out of every modeling decision.

The Semantic Gap



Handcrafted features can only cross this gap where meaning happens to correlate with the look, and those correlations break on new data: an orange-clad protest becomes "King's Day".

Next, Part 2: features that represent content. Networks that learn their own features (module 05) and features borrowed from networks trained on millions of images (module 06), scored on the same 40 images.

In the Notebook

- A four-variable codebook, and the hard cases it needs rules for
- Coding 40 images with a labeling tool in the notebook
- Recoding 20 of them: stability with kappa and alpha
- Category labels against your own codes
- Logistic regression and a random forest on handcrafted features
- Testing against your codes and reading the mistakes

part1_pixels/04_coding_images_and_classic_classifiers/notebooks/04_coding_images_and_classic_classifiers_exercises.ipynb · 6 exercises · CPU runtime · about 20 minutes of coding